

Exercise in
Monetary Transmission and Propagation Mechanism
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You are requested to investigate whether the Federal's Reserve policy affects core macroeconomic variables such as real GDP, and inflation. In US, the fed funds rate (ffr) is the variable utilized by the FR to conduct monetary policy. You will use the SVAR framework to examine if the monetary policy plays a key role that affects social welfare, as this is measured by the GDP and inflation.

Remarks:

1. To choose the optimal order p for the VAR use information criteria, such as BIC. You can search for p on $\{1, 2, 3\}$.
2. To measure the effect of monetary policy on the GDP and inflation use graphs of the IRF.
3. Use triangular the identification scheme for the shocks. You need to assume that the shocks of monetary policy affect both the GDP and the inflation, with a lag (last in the the cholesky decomposition).
4. To download the data use the <https://fred.stlouisfed.org>
5. For the analysis use the real chained-2009 GDP (GDPC96), a chain type price indexed at 2009 (gdpctpi) and the ffr (FEDFUNDS). All series should be transformed into quarter on quarter differences - log differences as follows. To this end, one can use the following transformations

$$\begin{aligned}Y_{1t} &= \ln \left(\frac{RGDP_t}{RGDP_{t-1}} \right) \times 100 \\Y_{2t} &= \ln \left(\frac{PI_t}{PI_{t-1}} \right) \times 100 \\Y_{3t} &= ffr_t\end{aligned}$$

Optimally, these should be seasonally adjusted.

At the end, you need to submit a small essay that presents the main research question, the theoretical model that you use to answer your research question, and the 3x3 graphs of the IRFs. Please also comment on the patterns that you observe. Moreover, please submit the R codes that implement the computations.

6. Small Extra Credit Opportunity:

In real-world applications, there is often substantial ambiguity regarding the true underlying model and the resulting impulse response functions (IRFs). To enhance the robustness of your analysis, apply cross-validation for VAR model selection. If the cross-validation selected model differs from the one selected by the information criterion (IC), compute the IRFs based on this alternative model as well. Compare the IRFs from both approaches and discuss any similarities

or differences. What do these results suggest about model uncertainty and its impact on structural inference?

References:

Primiceri, G. E. (2005). Time varying structural vector autoregressions and monetary policy. *The Review of Economic Studies*, 72(3), pp.821-852.

Stock, J. H., & Watson, M. W. (2001). Vector autoregressions. *Journal of Economic perspectives*, 101-115.